

TSMAl ex5

```
In [123... import numpy as np
import matplotlib.pyplot as plt
import sympy as sym
from scipy.optimize import root_scalar
```

Task 1.

1.1.

```
In [124... def find_all_roots(func, func_prime, args, search_range=(-10, 10), n_guesses=20):
    roots = set()
    for guess in np.linspace(search_range[0], search_range[1], n_guesses):
        try:
            result = root_scalar(
                func,
                fprime=func_prime,
                args=args, # Tuple of (w_val, theta_val)
                x0=float(guess),
                method='newton'
            )
            if result.converged:
                roots.add(round(result.root, 8))
        except:
            continue
    if not result.converged:
        #print("#####")
        x_guesses = np.linspace(search_range[0], search_range[1], n_guesses)
        for i, guess in enumerate(x_guesses):
            # Create a small bracket around the guess point
            left = max(search_range[0], guess - 1.0) # 1.0 is bracket width
            right = min(search_range[1], guess + 1.0)

            # Verify sign change (required for brentq)
            f_left = func(left, *args)
            f_right = func(right, *args)

            if f_left * f_right < 0: # Valid bracket
                try:
                    result = root_scalar(
                        func,
                        args=args,
                        bracket=[left, right],
                        method='brentq',
                        xtol=maxtol,
                        rtol=maxtol
                    )
                    if result.converged:
                        roots.add(round(result.root, 8))
                except:
                    continue

    return np.array(sorted(roots)), result.converged
```

In [179...

```

class Map:
    def __init__(self, expr, params, order): # dynamic variables first
        self.params = params
        self.expr = expr
        self.lhs = expr.lhs
        self.func = sym.lambdify((params), self.lhs, modules=['numpy'])
        self.order_tupl = tuple(order.split())
        self.fp_vals = 'call FP method first'

        self.prime_expr = sym.diff(self.lhs, params[0])
        self.prime_expr_func = sym.lambdify((params[0], *self.params[1:]), s

    def __call__(self, param_vals: dict):

        param_val_tupl = tuple(param_vals[key] for key in self.order_tupl)

        return self.func(*param_val_tupl)

    def FP_num(self, initial_guess=0.0, default_params=None, plot=False, maxtol=
        x = self.params[0]
        FP_eq = sym.Eq(x, self.lhs)

        g = FP_eq.lhs - FP_eq.rhs
        fp_func = sym.lambdify((x, *self.params[1:]), g, modules=['numpy'])
        prime_func = sym.lambdify((x, *self.params[1:]), sym.diff(g, x), mod

        args = [default_params[key] for key in self.order_tupl]
        args = tuple(args[1:])

        roots, converged = find_all_roots(fp_func, prime_func, args, search_ran

        # result = root_scalar(
        #     fp_func,
        #     fprime=prime_func,
        #     args=args,
        #     x0=float(initial_guess),
        #     method='newton'
        # )

        if converged:
            self.fp_vals = roots

            if plot:
                x_arr = np.linspace(-10, 10, 100)
                plt.plot(x_arr, self(default_params), label='map')
                plt.plot(x_arr, x_arr, label='bisectrix')
                plt.scatter(roots, roots, label='FP', color='g')
                plt.title('expression return plot')
                plt.xlabel('x_{n}')
                plt.ylabel('x_{n+1}')
                plt.legend()
                plt.show()

            return roots
        else:
            print('did not converge')
            print("here is a plot: ")
            x_arr = np.linspace(-10, 10, 100)
            plt.plot(x_arr, self(default_params))

```

```

plt.plot(x_arr,x_arr)
plt.title('expression return plot')
plt.show()

def FP(self, initial_guess=0.0, default_params=None, plot=False, maxtol=1e-4):
    x = self.params[0]
    FP_eq = sym.Eq(x, self.lhs)

    try:
        fp_solutions = sym.solve(FP_eq, x)

        if fp_solutions:
            other_params = [p for p in self.params if p != x]
            self.FP_func = sym.lambdify(other_params, fp_solutions[0], n

            return fp_solutions
    except Exception as e:
        print(f"Could not find fixed point symbolically: {e}")

    roots = self.FP_num(initial_guess=initial_guess, default_params=c

    return roots

return None

def stability(self, param_vals, prints=False):
    if type(self.fp_vals) == str:
        print(self.fp_vals)

    param_val_tupl = tuple(param_vals[key] for key in self.order_tupl)
    #print(param_val_tupl)

    local_slope = self.prime_expr_func(*param_val_tupl)
    abs_slope = abs(local_slope)

    stab_list = []

    for i, slope in enumerate(abs_slope):
        if prints:
            if slope > 1:
                print(f'Unstable fp because local slope is |{local_slope|
            elif slope < 1:
                print(f'Stable because local slope is |{local_slope| <
            else:
                print(f'hihger order analysis needed because local slope
        stab_list.append(slope < 1) # True stable, False unstable

    return local_slope, stab_list

def k_iterate(self, k, simpl=False):
    x_current = self.params[0] # default
    for _ in range(k):
        x_current = self.lhs.subs(self.params[0], x_current)
    if simpl:
        return sym.simplify(x_current), sym.lambdify(self.params, x_curr
    else:
        return x_current, sym.lambdify(self.params, x_current, 'numpy')

```

```
In [126... x,w,theta = sym.symbols('x w theta')

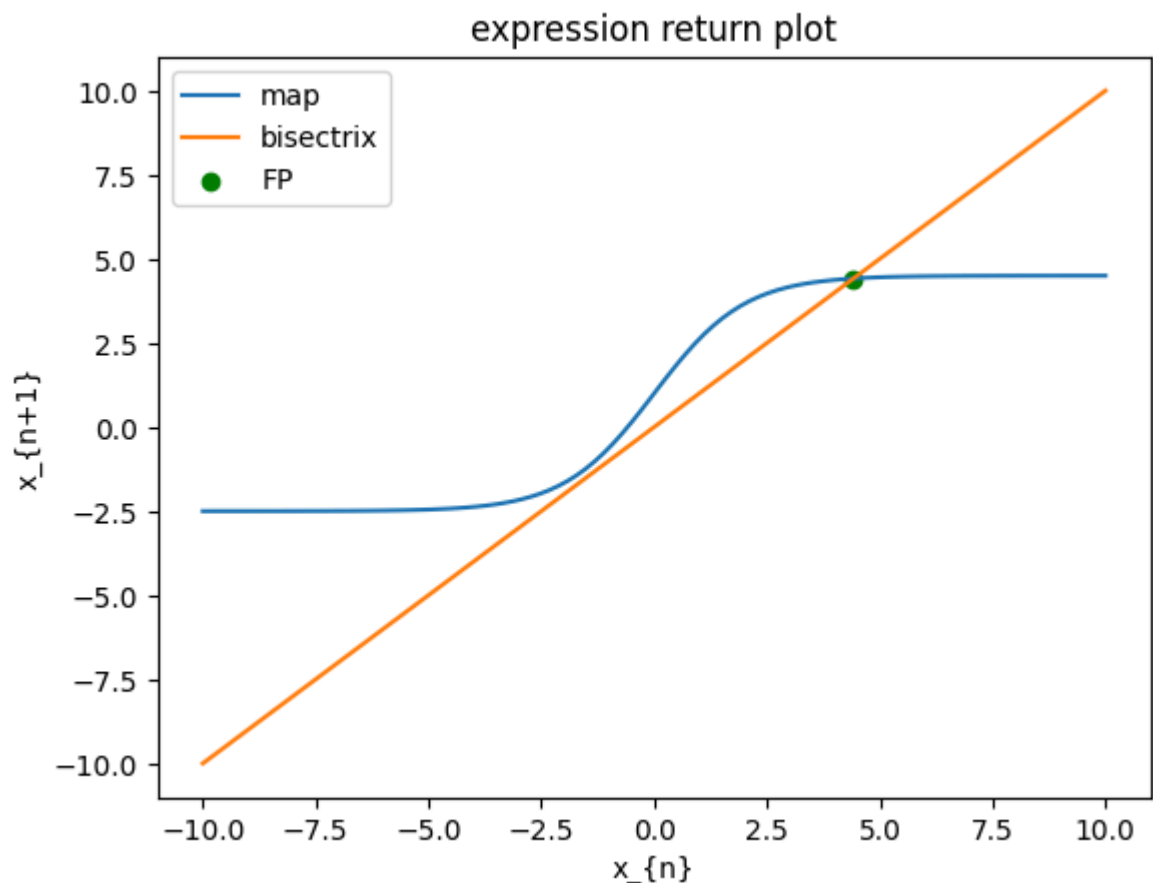
expr = sym.Eq(w/(1+sym.exp(-x))+theta, 0) # lhs,rhs

# a = expr.free_symbols # set of symbols

MAP_task1 = Map(expr, [x,w,theta], 'x w theta')
FPexpr = MAP_task1.FP(initial_guess=4.5,default_params={'x': np.linspace(-10,10,100)})
print(MAP_task1.fp_vals)

FP_slope = MAP_task1.stability({'x': MAP_task1.fp_vals, 'w':7, 'theta':-2.5})

Could not find fixed point symbolically: multiple generators [x, exp(x)]
No algorithms are implemented to solve equation x + (-theta - w/(1 + exp(-x)))
```



[4.41647135]

Stable because local slope is $|[0.08253193]| < 1$

We find the FP numerically and the stability which is stable bc the slope is less than 1.

1.2

```
In [127... from tqdm import tqdm

bifur_MAP = Map(expr, [x,w,theta], 'x w theta')

theta_range = np.linspace(-10,0,1000)

FP_vals = []
theta_vals = []
colors = []

import warnings
```

```

with warnings.catch_warnings():
    warnings.simplefilter("ignore", RuntimeWarning)
    for theta_val in tqdm(theta_range):
        FPexpr = MAP_task1.FP_num(initial_guess=0, default_params={'x': np.li

        if FPexpr is not None:
            theta = np.ones(len(FPexpr))*theta_val
            slope, stab_list = MAP_task1.stability({'x': MAP_task1.fp_vals,
            color = ['g' if stab else 'r' for stab in stab_list]
            FP_vals.extend(FPexpr)
            theta_vals.extend(theta)
            colors.extend(color)

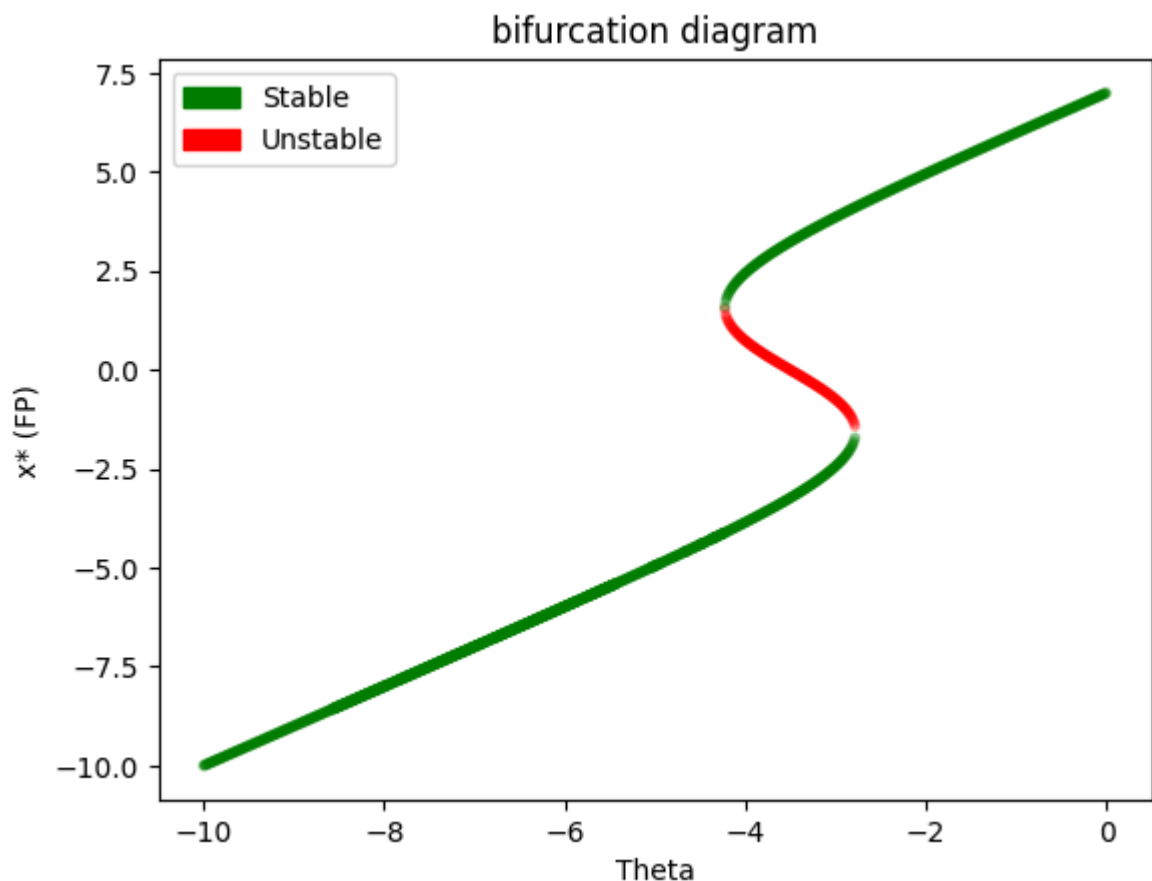
```

100%|██████████| 1000/1000 [00:40<00:00, 24.94it/s]

```

In [128... import matplotlib.patches as mpatches
plt.scatter(theta_vals, FP_vals, color=colors, alpha=0.3, marker='.')
plt.title('bifurcation diagram')
plt.xlabel('Theta')
plt.ylabel('x* (FP)')
green_patch = mpatches.Patch(color='g', label='Stable')
red_patch = mpatches.Patch(color='r', label='Unstable')
plt.legend(handles=[green_patch, red_patch])
plt.show()

```



We go from one stable FP to one stable and one halfstable, then to three FP, two stable one unstable and then one stable again. The transitions are bifurcations.

Task 2

2.2

```
In [180... x,r = sym.symbols('x r')
logis_eq = sym.Eq(r*x*(1-x), 0)
```

```
In [181... logistic_map = Map(logis_eq,[x,r],'x r')
```

```
In [151... def plot_map(map,param_dict,iter,Title=None):
    #x_vals = np.zeros(iter)
    fig, ax = plt.subplots(figsize=(12, 6))
    x = param_dict['x']
    x0 = x
    x_plot_vals = [x,0]
    x_min = x0
    x_max = x0
    for i in range(iter):
        if i == 0:
            x = map(param_dict)
            param_dict['x'] = x
            x_plot_vals[1] = x

            ax.plot([x_plot_vals[0],x_plot_vals[1]],[x_plot_vals[1],x_plot_v
            if x < x_min:
                x_min = x
            if x > x_max:
                x_max = x
        else:
            ax.plot([x_plot_vals[0],x_plot_vals[1]],[x_plot_vals[1],x_plot_v
            x_plot_vals[0] = x_plot_vals[1]
            x = map(param_dict)
            param_dict['x'] = x

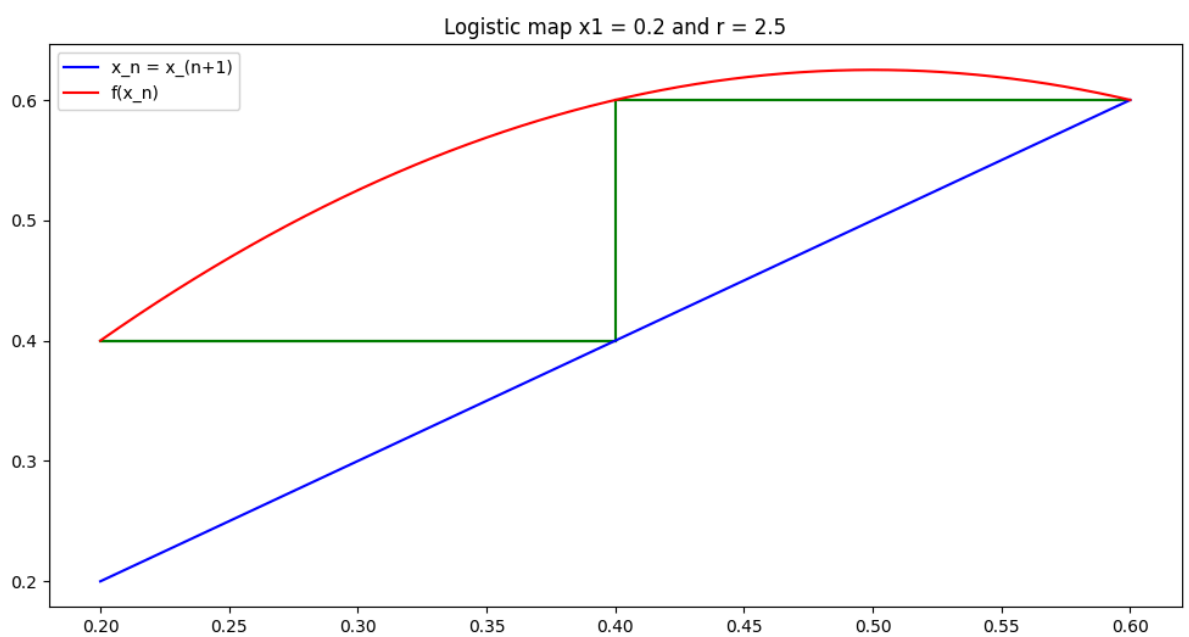
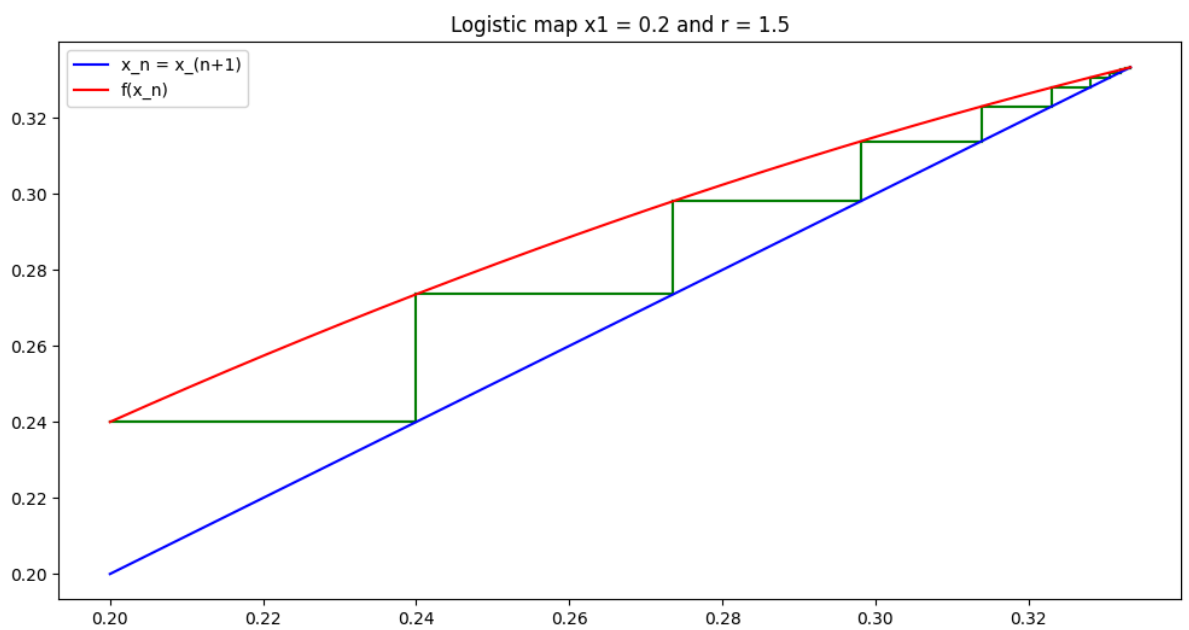
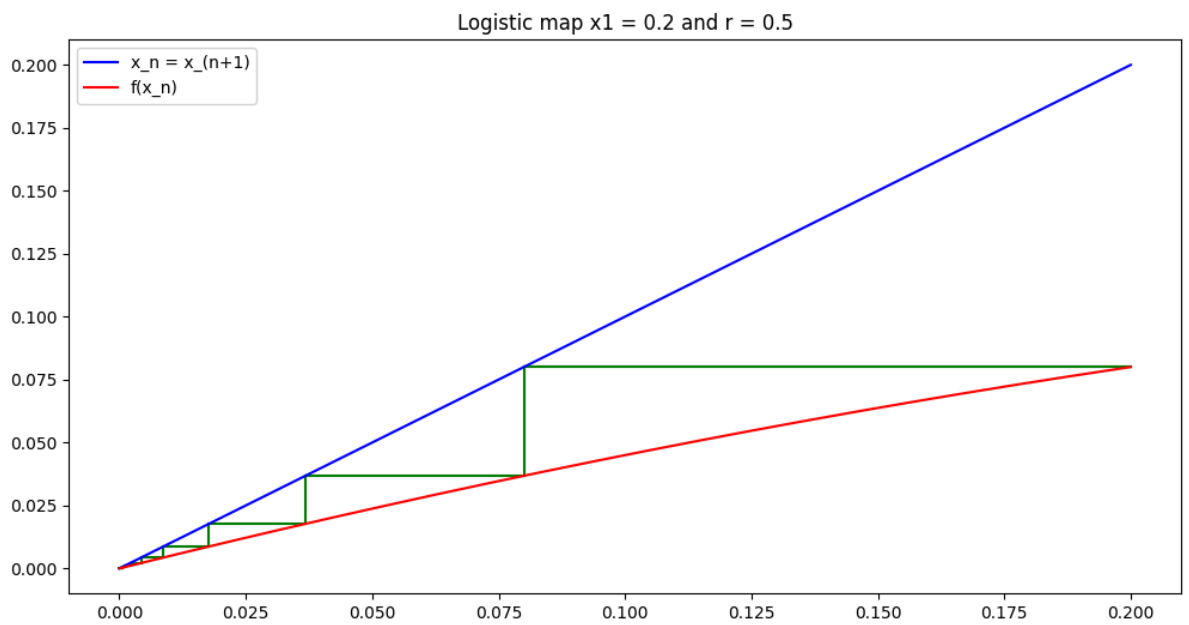
            x_plot_vals[1] = x
            ax.plot([x_plot_vals[0],x_plot_vals[0]],[x_plot_vals[0],x_plot_v
            if x < x_min:
                x_min = x
            if x > x_max:
                x_max = x

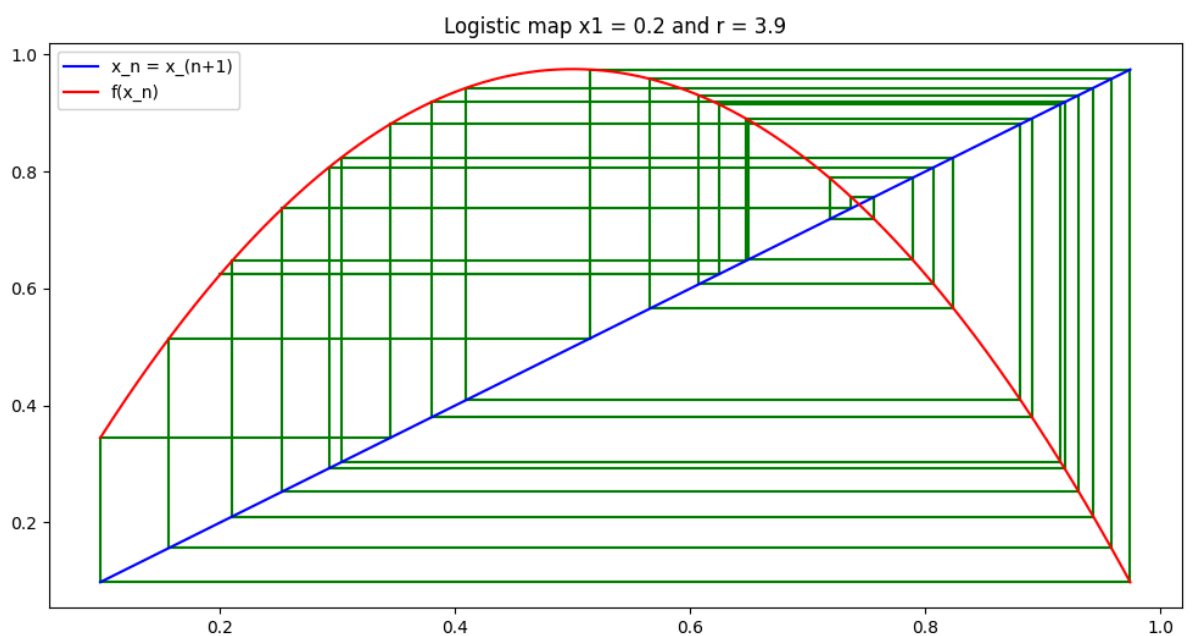
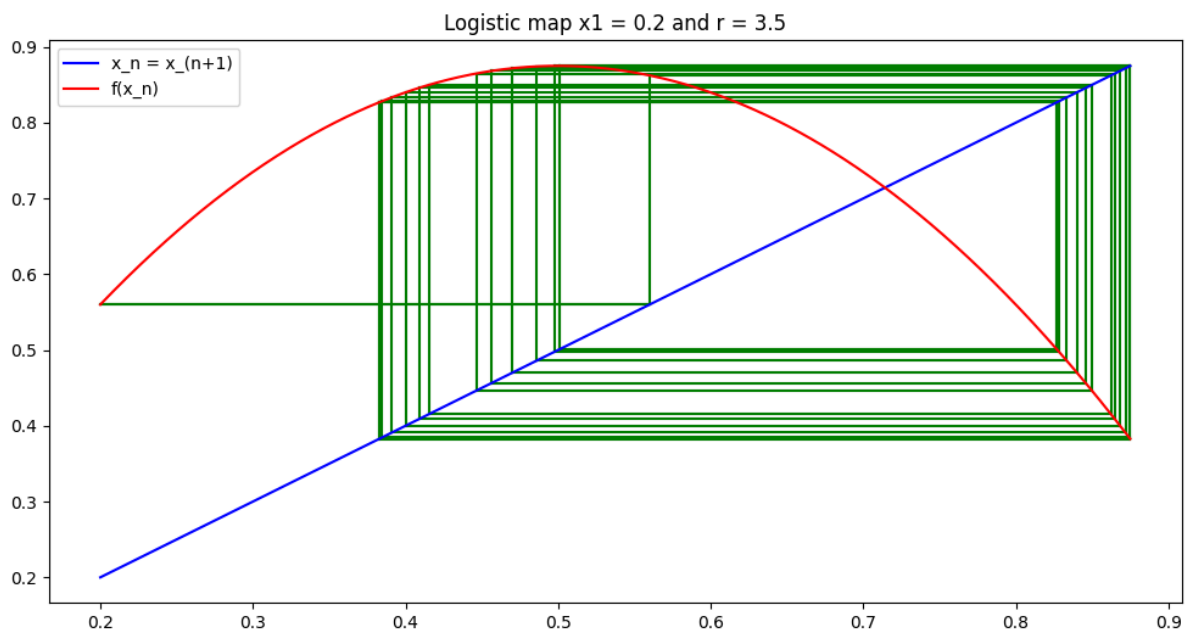
    x_plot = np.linspace(x_min,x_max,1000)
    ax.plot(x_plot,x_plot,label=f"x_n = x_(n+1)",color="blue")
    param_dict['x'] = x_plot
    ax.plot(x_plot,map(param_dict),label=f"f(x_n)",color="red")
    ax.legend()
    ax.set_title(Title)

    fig.show()
```

```
In [152... x0 = 0.2
r = [0.5, 1.5, 2.5, 3.5, 3.9]
for R in r:
    plot_map(logistic_map,{'x': x0,'r': R},30,Title=f'Logistic map x1 = {x0
```

```
/var/folders/pd/2bgmf1pd7fv79_2hltggtpsw0000gn/T/ipykernel_54592/339233771
2.py:41: UserWarning: Matplotlib is currently using module://matplotlib_inl
ine.backend_inline, which is a non-GUI backend, so cannot show the figure.
    fig.show()
```





2.4

```
In [173... rand_x0 = np.random.rand(1000)

r_vals = np.arange(0,4,0.001)

len_traj = 100

dum_shape = np.ones(len(rand_x0))

x_plot = []
r_plot = []

for r in r_vals:
    param_dict = {'x':rand_x0,'r':r}
    for i in range(len_traj):
        x = logistic_map(param_dict)
        param_dict['x'] = x

    x_plot.extend(x)
```



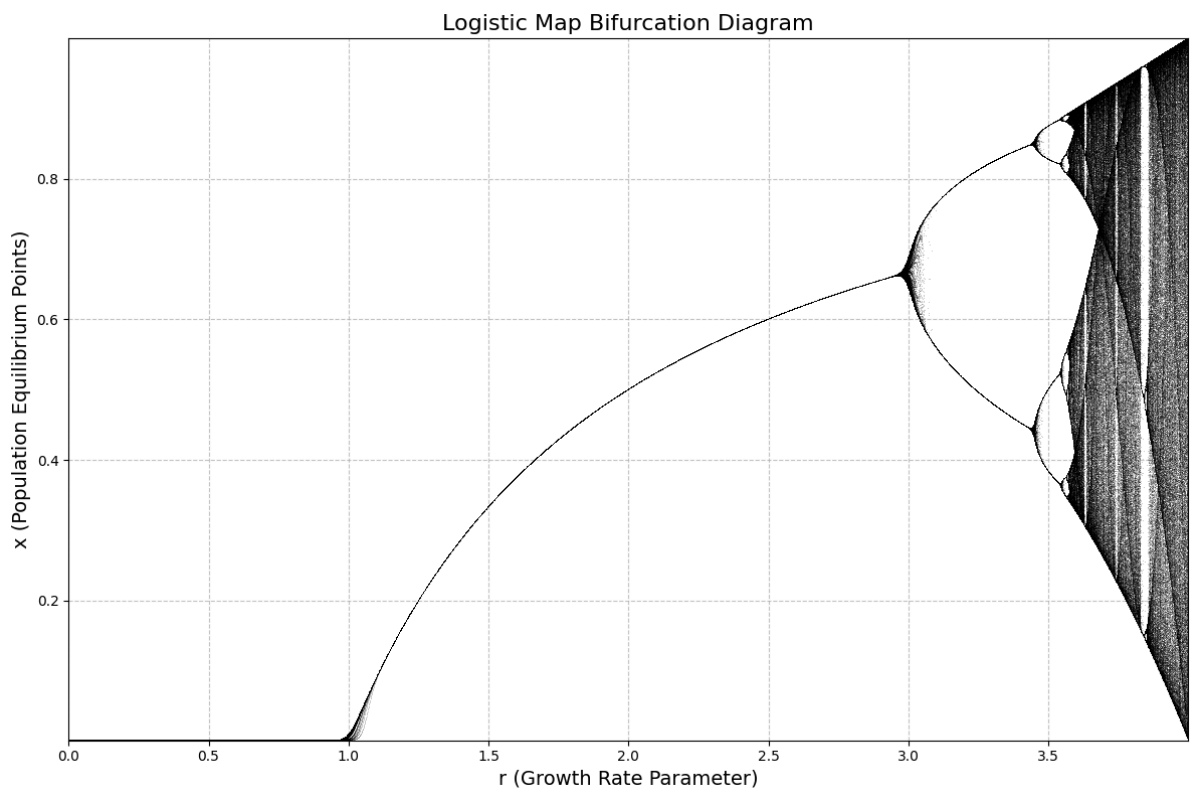
```

r_plot.extend(r*dum_shape)
#print(r)

plt.figure(figsize=(12,8))
plt.plot(r_plot, x_plot, ',', alpha=0.2, color='black')

plt.title("Logistic Map Bifurcation Diagram", fontsize=16)
plt.xlabel("r (Growth Rate Parameter)", fontsize=14)
plt.ylabel("x (Population Equilibrium Points)", fontsize=14)
plt.xlim(r_vals.min(), r_vals.max())
plt.ylim(x.min(), x.max())
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()

```



2.5

```

In [212... rand_x0 = np.random.rand(1000)

r_vals = np.arange(3.44500,3.57000,0.00001)

len_traj = 100

dum_shape = np.ones(len(rand_x0))
print(len(r_vals))

x_plot = []
r_plot = []

for r in r_vals:
    param_dict = {'x':rand_x0,'r':r}
    for i in range(len_traj):
        x = logistic_map(param_dict)
        param_dict['x'] = x

    x_plot.extend(x)
    r_plot.extend(r*dum_shape)

```

```

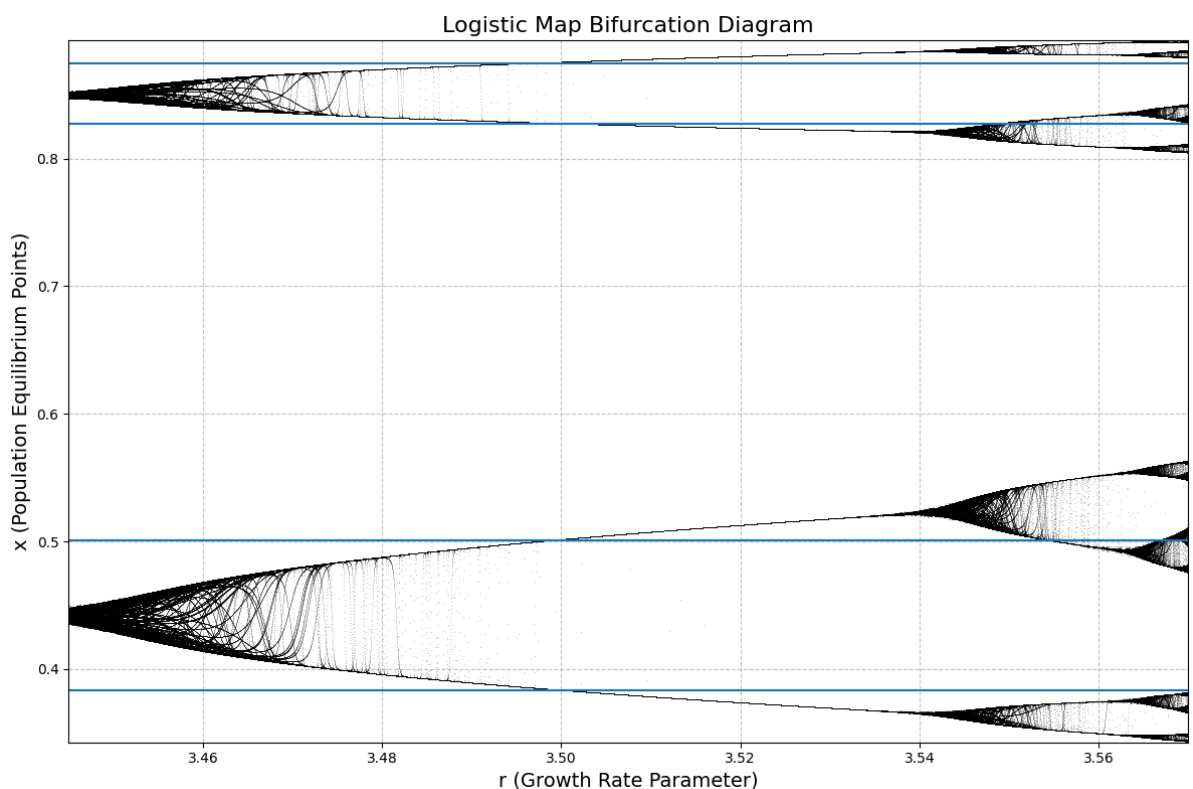
# print(r)

plt.figure(figsize=(12,8))
plt.plot(r_plot, x_plot, ',', alpha=0.2, color='black')

plt.title("Logistic Map Bifurcation Diagram", fontsize=16)
plt.xlabel("r (Growth Rate Parameter)", fontsize=14)
plt.ylabel("x (Population Equilibrium Points)", fontsize=14)
plt.xlim(r_vals.min(), r_vals.max())
plt.ylim(x.min(), x.max())
plt.grid(True, linestyle='--', alpha=0.7)
plt.axhline(y=0.8275)
plt.axhline(y=0.501)
plt.axhline(y=0.875)
plt.axhline(y=0.383)
plt.tight_layout()
plt.show()

```

12500



The horizontal lines are used below to check the cycle.

Wibe coded (using Gemini 2.5 pro) a C version for some speed, and let the map iterate 5000 times per 1000 r samples such that values have a lot of time to converge to attractor and we get some cleaner plots. Got a bit carried away with the plots:)

```

In [ ]: import pandas as pd

def plot_bifur_from_c(file, save=False, alpha=0.25, figsize=(12, 8)):
    # Read the bifurcation data
    try:
        df = pd.read_csv(file)
    except FileNotFoundError:
        print("Error: bifurcation_data.csv not found. Run the C program first")
        exit()

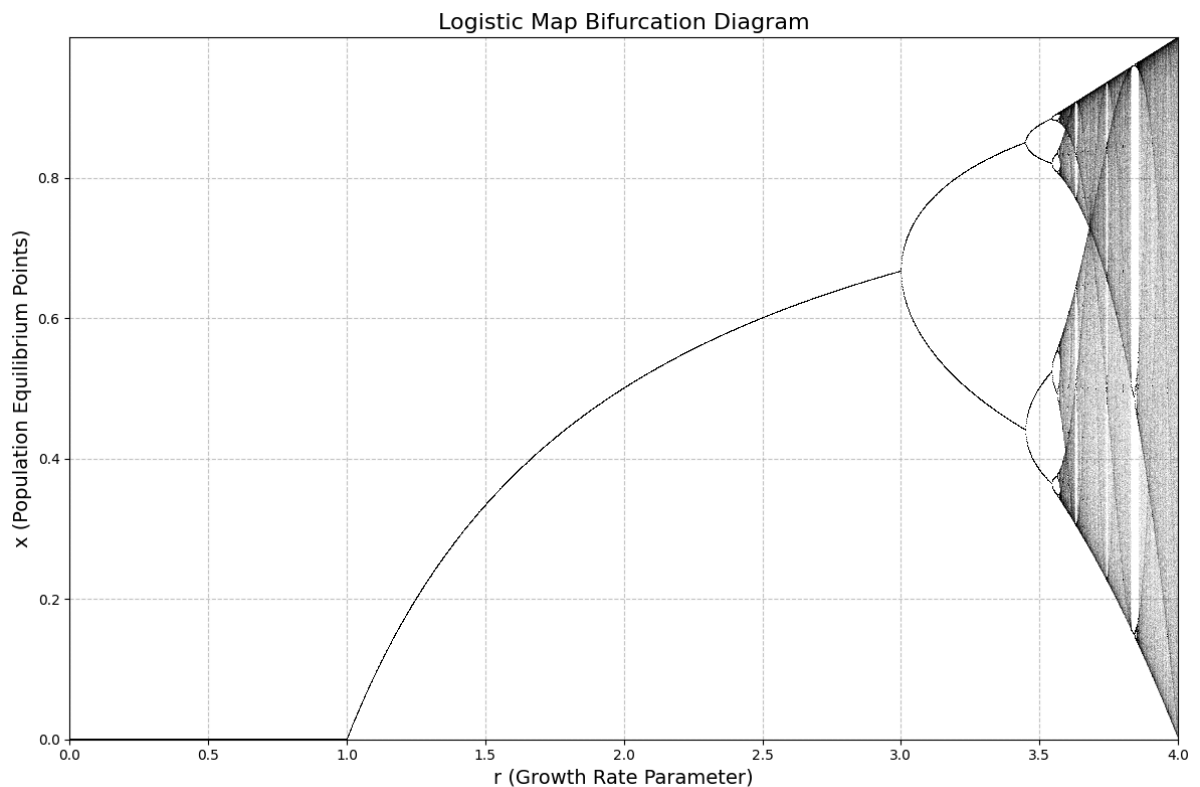
    # Plotting

```

```
plt.figure(figsize=figsize)
plt.plot(df['r'], df['x'], ',', alpha=alpha, color='black') # Use ',' for

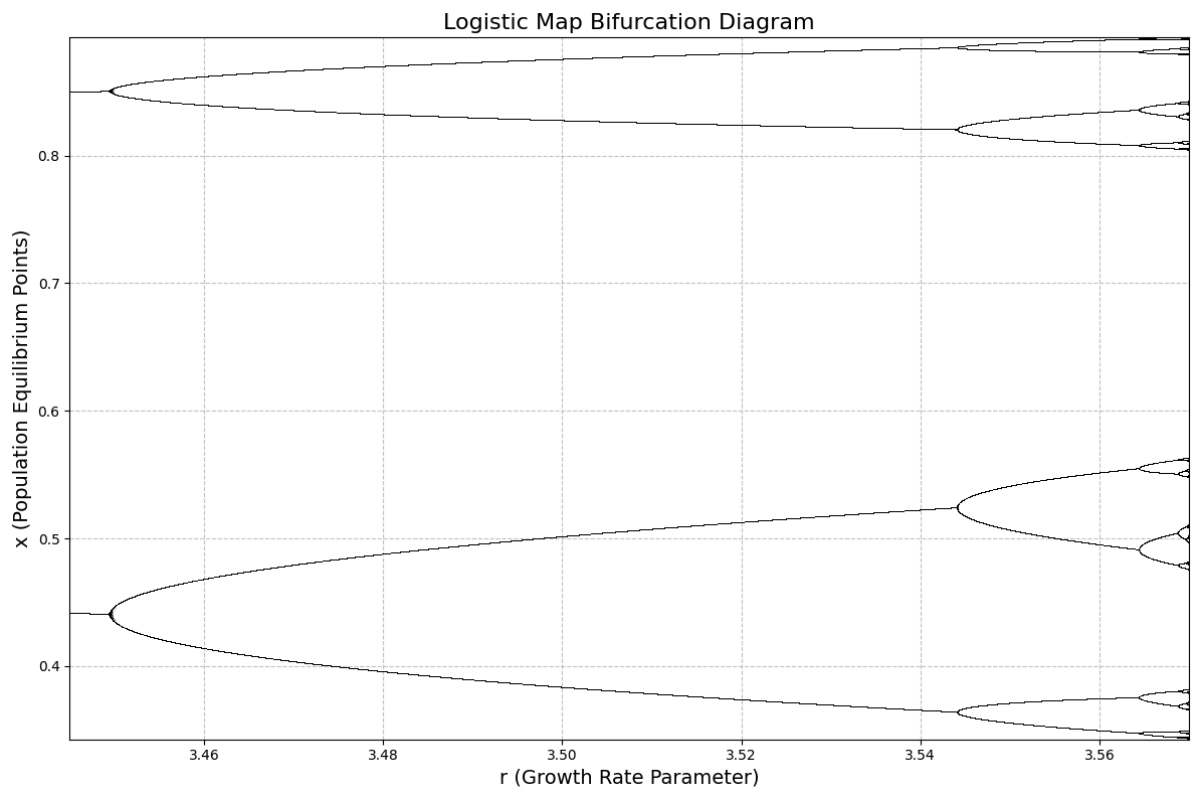
plt.title("Logistic Map Bifurcation Diagram", fontsize=16)
plt.xlabel("r (Growth Rate Parameter)", fontsize=14)
plt.ylabel("x (Population Equilibrium Points)", fontsize=14)
plt.xlim(df['r'].min(), df['r'].max())
plt.ylim(df['x'].min(), df['x'].max())
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
if save:
    save_name = file.rsplit('.', 1)[0] + '.png'
    plt.savefig(save_name, dpi=300)
    print(f"Bifurcation diagram plotted and saved as {save_name}")
plt.show()
```

In [130... plot_bifur_from_c("bifurcation_data.csv", alpha=0.05)



2.5

In [131... plot_bifur_from_c("endpoints_data_mpfr.csv")



2.5 still

We observe 4 branches for $r \approx 3.5$ so there should be a $4 > p = 3$ cycle. We verify mathematically by using sympy to create a $f^4(x)$ map and verify if $f^4(x) = x$

```
In [220... k3expr, k3logistic_func = logistic_map.k_iterate(4)

print("checking the 4 cycle with values found visually for r = 3.5")
print(f'f^4(0.383) = {k3logistic_func(x=0.383,r=3.5)}')
print(f'f^4(0.501) = {k3logistic_func(x=0.501,r=3.5)}')
print(f'f^4(0.8275) = {k3logistic_func(x=0.8275,r=3.5)}')
print(f'f^4(0.875) = {k3logistic_func(x=0.875,r=3.5)}')

print("also checking with iterativly applying map")
x0 = 0.383
param_dict = {'x':x0, 'r':3.5}
for i in range(12):
    x = logistic_map(param_dict)
    param_dict['x'] = x
    print(round(x,3))
print('we clearly see a cycle of order 4, since 4 distict values repeat in a
```

checking the 4 cycle with values found visually for $r = 3.5$

$f^4(0.383) = 0.3828152378945099$

$f^4(0.501) = 0.5008804475158373$

$f^4(0.8275) = 0.826936001536345$

$f^4(0.875) = 0.87499717950388$

also checking with iteratively applying map

0.827

0.501

0.875

0.383

0.827

0.501

0.875

0.383

0.827

0.501

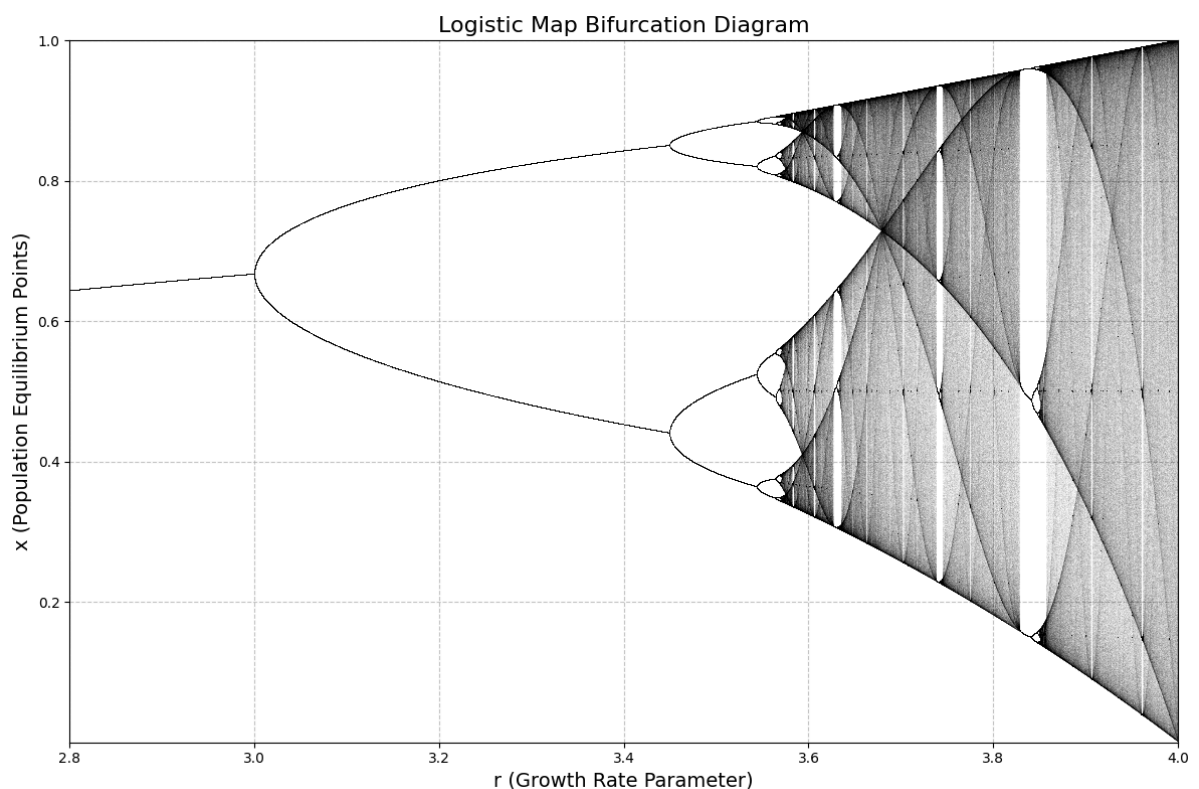
0.875

0.383

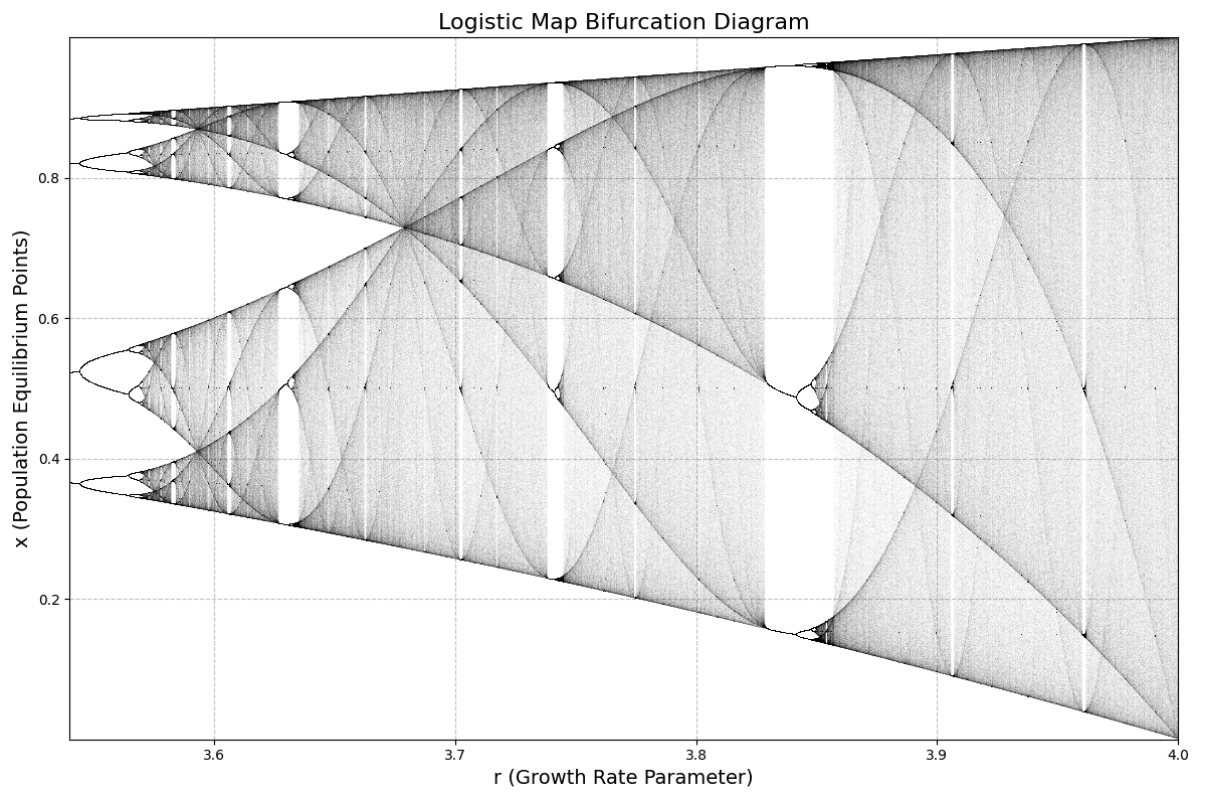
we clearly see a cycle of order 4, since 4 distinct values repeat in a cyclic manner.

some more pretty plots

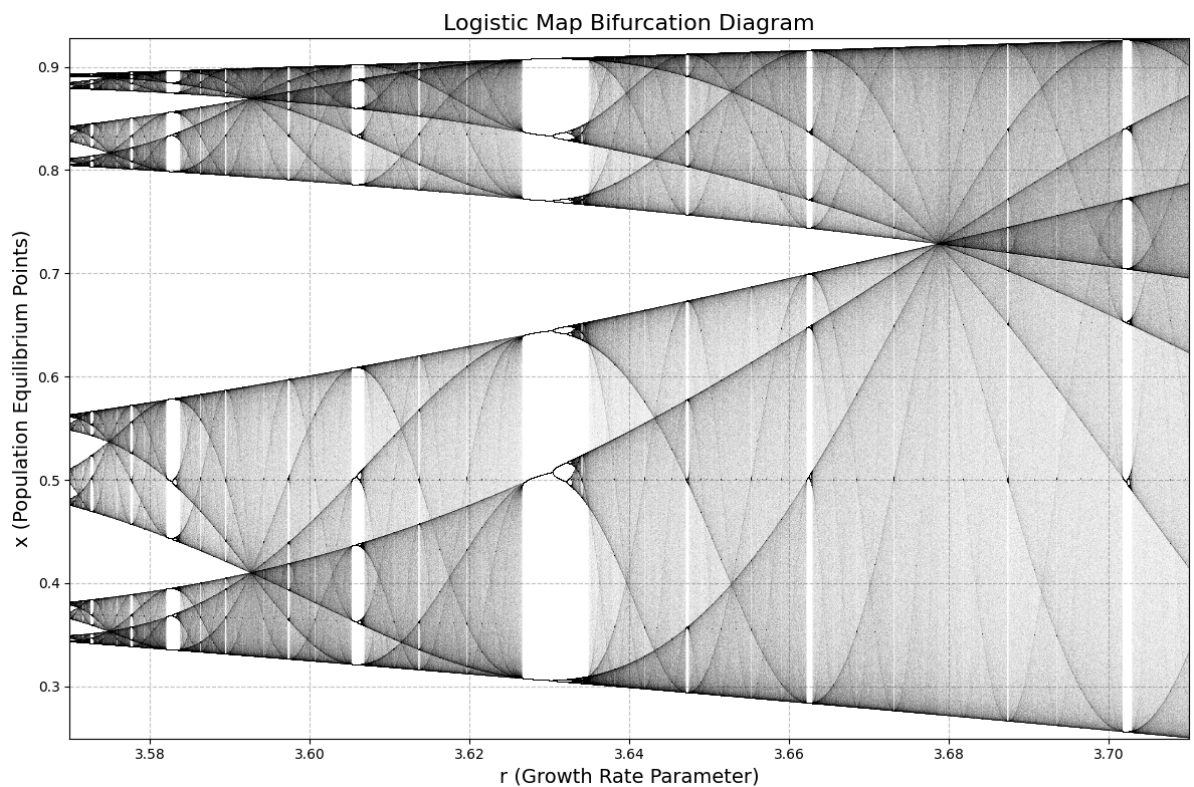
```
In [132... plot_bifur_from_c("endpoints_data.csv",alpha=0.02)
```



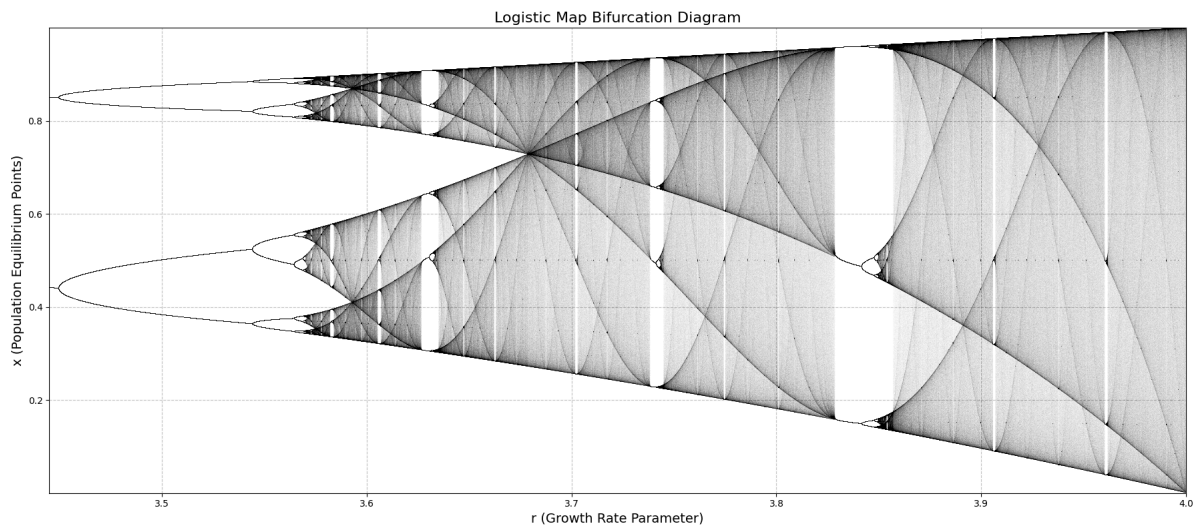
```
In [133... plot_bifur_from_c("endpoints_data_3_54_4.csv",alpha=0.02)
```



```
In [134... plot_bifur_from_c("endpoints_data_3_57_71.csv",alpha=0.01)
```



```
In [135... plot_bifur_from_c("endpoints_data_3_445_4.csv",alpha=0.008,figsize=(18,8))
```

2.6

```
In [ ]: logistic_map_func = logistic_map.func
x = np.random.rand(1000)
for i in range(4000):
    x = logistic_map_func(x, r=3.5)

print(x.shape)

(1000,)
```

```
In [157...] def find_distinct(sorted_arr, accuracy):
    """Count distinct values."""
    if len(sorted_arr) == 0:
        return 0
    distinct = 1
    current = sorted_arr[0]
    for val in sorted_arr[1:]:
        if abs(val - current) > max(accuracy * abs(current), accuracy):
            distinct += 1
            current = val
    return distinct

def find_bifur(r0, map_func, max_bifur_search, filename,
               warmup_len=10000,
               analyse_traj_len=5000,
               initial_min_r_increment=1e-5,
               accuracy_distinct=1e-10,
               tolerance_decay=0.5,
               min_dist_bifur=1e-5,
               min_r_increment_limit=1e-12,
               r_high_init=4
               ):

    bifur_points = []
    r_current = r0
    min_r_increment = max(initial_min_r_increment, min_r_increment_limit)

    while len(bifur_points) < max_bifur_search:
        # Bisection setup
        r_low = r_current
        r_high = r_high_init
        valid_bifur = False
        attempts = 0
```

```

while not valid_bifur and attempts < 10:
    # Get initial distinct count at r_low
    x = np.random.rand()
    for _ in range(warmup_len):
        x = map_func(np.float128(x), np.float128(r_low))
    x_vals = np.zeros(analyse_traj_len, dtype=np.float128)
    for i in range(analyse_traj_len):
        x = map_func(np.float128(x), np.float128(r_low))
        x_vals[i] = x
    n_low = find_distinct(np.sort(x_vals), accuracy_distinct)

    # Bisection loop
    while r_high - r_low > min_r_increment:
        r_mid = (r_low + r_high) / 2.0

        # Compute distinct count at midpoint
        x = np.random.rand()
        for _ in range(warmup_len):
            x = map_func(np.float128(x), np.float128(r_mid))
        x_vals = np.zeros(analyse_traj_len, dtype=np.float128)
        for i in range(analyse_traj_len):
            x = map_func(np.float128(x), np.float128(r_mid))
            x_vals[i] = x
        n_mid = find_distinct(np.sort(x_vals), accuracy_distinct)

        # Narrow interval
        if n_mid > n_low:
            r_high = r_mid
        else:
            r_low = r_mid

    candidate_point = np.float128(r_mid)

    # Enforce minimum distance and precision limits
    if bifur_points:
        if candidate_point != r_high_init:
            prev_point = bifur_points[-1]
            distance = abs(candidate_point - prev_point)

            if distance < min_dist_bifur:
                print(f"Adjusting precision: {distance:.2e} < {min_dist_bifur:.2e}")
                min_r_increment = max(min_r_increment * 0.1, min_r_increment)
                attempts += 1
                r_low = prev_point
                r_high = r_high_init
                continue

    valid_bifur = True

if not valid_bifur:
    r_low = prev_point

# Store valid bifurcation
bifur_points.append(candidate_point)
write_value_to_file(candidate_point, filename)
print(f"Found bifurcation {len(bifur_points)} at {candidate_point:.12e}")

# Update search parameters with decay limits
if len(bifur_points) > 1:
    prev_interval = bifur_points[-1] - bifur_points[-2]

```



```

        min_r_increment = max(
            prev_interval * tolerance_decay,
            min_r_increment_limit
        )
    else:
        min_r_increment = max(
            initial_min_r_increment,
            min_r_increment_limit
        )

    r_current = candidate_point

    return bifur_points

def write_value_to_file(value, filename):
    with open(filename, 'a') as f:
        f.write(f"{value}\n")

def read_values_from_file(filename):
    return np.loadtxt(filename)

```

The code is an attempt at implementing a bisection method for finding qualitative differences in amount of unique points in the trajectory after it has been warmed up for a while to reach the attractor. It works by having an interval of r values, then calculate the amount of distinct values at the low and mid point of the interval, if they are the same an bifurcation comes after the midpoint, and one sets the new low point at the midle and repeat, other wise if the mid point and the low point have different distinct points then a bifurcation happened before the point, and the midpoint is choosen as the high point. This continues until the interval is small, in this case smaller than a set `min_r_increment` then the midpoint is just choosen as the bifurcation point and we repeat our search with the low point at the found bifur point. To increase accuracy we could have used this method for making the interval in which the bifurcation is small then goint from the low end of the interval and incrementally increase r with a small increment such that the bifurcation point can be estimated better.

```

In [ ]: r0 = 3.44
        filename = "bifurpoints.txt"

        bifurcations = find_bifur(
            r0=r0,
            map_func=logistic_map_func,
            max_bifur_search=15, # Find 10 bifurcations
            filename="bifurcations.txt",
            warmup_len=10000,
            analyse_traj_len=5000,
            initial_min_r_increment=1e-10,
            accuracy_distinct=1e-12,
            tolerance_decay=0.1,
            min_dist_bifur=1e-13,
            min_r_increment_limit=1e-12
        )

```

```

Found bifurcation 1 at 3.448533697734
Found bifurcation 2 at 3.448533799874
Found bifurcation 3 at 3.448572758979
Found bifurcation 4 at 3.543704812260
Found bifurcation 5 at 3.565093649185
Found bifurcation 6 at 3.570190207984
Found bifurcation 7 at 3.999580263875
Adjusting precision: 0.00e+00 < 1e-13
Adjusting precision: 0.00e+00 < 1e-13
Adjusting precision: 0.00e+00 < 1e-13
Found bifurcation 8 at 3.999973766492
Adjusting precision: 0.00e+00 < 1e-13
Found bifurcation 9 at 3.999996720812
Found bifurcation 10 at 3.999998360406
Found bifurcation 11 at 3.99999897525
Adjusting precision: 0.00e+00 < 1e-13
Found bifurcation 12 at 3.99999987191
Found bifurcation 13 at 3.99999993595
Found bifurcation 14 at 3.99999999600
Adjusting precision: 0.00e+00 < 1e-13
Found bifurcation 15 at 3.99999999950

```

The code runs all the way to the end and most likely finds some points that are the same bifurcation, it also struggles to find something beyond 3.56, which from looking at the bifurcation diagrams is where the bifurcation points are really close together.

2.7

```

In [146... def estimate_feigenbaum(bifurcation_points):
    ratios = []
    for n in range(1, len(bifurcation_points)-1):
        dr_n = bifurcation_points[n] - bifurcation_points[n-1]
        dr_n1 = bifurcation_points[n+1] - bifurcation_points[n]
        d_n = dr_n / dr_n1
        ratios.append(d_n)
    return ratios

valid_bifur_r = [3.448, 3.543, 3.564, 3.570] # picked from file written in e
feigenbaum_est = estimate_feigenbaum(valid_bifur_r)
print(feigenbaum_est)

[4.523809523809553, 3.500000000000111]

```

We expect to approximate the feigenbaum constant if we find the correct bifurcation points. From wibe coded c code using MPFR library for increased floating point accuracy we got the output:

Finding r_0 (1->2 cycles)...

r_0 (onset of 2-cycle): 2.9989263212845260214592827572543923577919

Finding r_1 (2->4 cycles)...

r_1 (onset of 4-cycle): 3.4490572480052443944697443872246495125654

Finding r_2 (4->8 cycles)...

$$\Delta_7 = (r_7 - r_6) / (r_8 - r_7) = 4.66920160910000000000000000000000$$

$$\Delta_8 = (r_8 - r_7) / (r_9 - r_8) = 4.66920160910000000000000000000000$$

Where we clearly see that our bifurcation points are not very accurately found, and that the delta values are supposed to approach the feigenbaum constant.

```
In [148... x, r = sym.symbols('x r')

sin_map_expr = sym.Eq(r*sym.sin(sym.pi*x), 0)

sin_map = Map(sin_map_expr, [x, r], 'x r')
```

```
In [ ]: rand_x0 = np.random.rand(1000)

r_vals = np.arange(0,1,0.0005)

len_traj = 1000

dum_shape = np.ones(len(rand_x0))

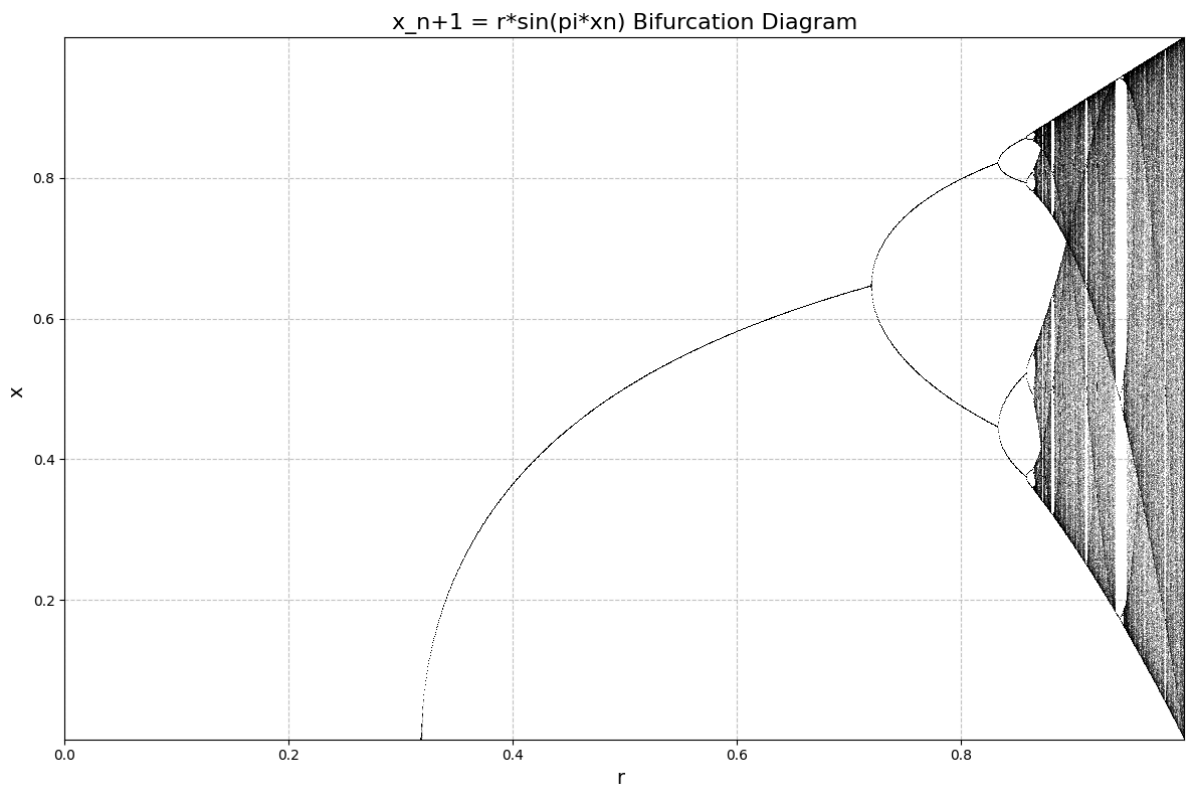
x_plot = []
r_plot = []

for r in r_vals:
    param_dict = {'x':rand_x0,'r':r}
    for i in range(len_traj):
        x = sin_map(param_dict)
        param_dict['x'] = x

    x_plot.extend(x)
    r_plot.extend(r*dum_shape)
    #print(r)

plt.figure(figsize=(12,8))
plt.plot(r_plot, x_plot, ',', alpha=0.2, color='black')

plt.title("x_{n+1} = r*sin(pi*x_n) Bifurcation Diagram", fontsize=16)
plt.xlabel("r ", fontsize=14)
plt.ylabel("x ", fontsize=14)
plt.xlim(r_vals.min(), r_vals.max())
plt.ylim(x.min(), x.max())
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
```



```
In [158... r0 = 0.82
filename = "bifurpoints2.txt"
sin_map_func = sin_map.func

bifurcations = find_bifur(
    r0=r0,
    map_func=sin_map_func,
    max_bifur_search=15, # Find 10 bifurcations
    filename="bifurcations.txt",
    warmup_len=10000,
    analyse_traj_len=5000,
    initial_min_r_increment=1e-10,
    accuracy_distinct=1e-12,
    tolerance_decay=0.1,
    min_dist_bifur=1e-13,
    min_r_increment_limit=1e-12,
    r_high_init=1
)
```

```
Found bifurcation 1 at 0.833024293585
Found bifurcation 2 at 0.858505239715
Found bifurcation 3 at 0.860716095344
Found bifurcation 4 at 0.864116581298
Found bifurcation 5 at 0.882428995146
Found bifurcation 6 at 0.945807114950
Found bifurcation 7 at 0.996612944684
Adjusting precision: 0.00e+00 < 1e-13
Found bifurcation 8 at 0.999576618086
Found bifurcation 9 at 0.999788309043
Found bifurcation 10 at 0.999986769315
Adjusting precision: 0.00e+00 < 1e-13
Found bifurcation 11 at 0.999998346164
Found bifurcation 12 at 0.999999173082
Found bifurcation 13 at 0.999999948318
Adjusting precision: 0.00e+00 < 1e-13
Found bifurcation 14 at 0.999999993540
Found bifurcation 15 at 0.999999996770
```

```
In [159... valid_bifur_r = [0.833, 0.858, 0.861, 0.864, 0.882, 0.945]

feigenbaum_est = estimate_feigenbaum(valid_bifur_r)
print(feigenbaum_est)

[8.333333333333334, 1.0, 0.16666666666666666, 0.2857142857142862]
```

I am guessing that we are supposed to recover the Feigenbaum constant, or approach some other constant for this system, however this we dont manage, probably bc of the finding bifurcation points.

1.11

$$\sigma(x) = \frac{1}{1+e^x}$$

$$x_{t+1} = f(x_t, \omega, \theta) = \omega \cdot \sigma(x_t) + \theta$$

fix pts . $x_{t+1}^* = x_t^* = x^*$

$$\frac{\omega}{1+e^{-x^*}} + \theta = x^*$$

$$\omega + \theta + e^{-x^*} = x^* + x^* e^{-x^*}$$

$$\omega + \theta = x^* + e^{-x^*} (x^* - 1)$$

$$\omega + \theta = x^* + x^* e^{-x^*} - e^{-x^*}$$

Numerical.

2. Logistic Map

$$x_{n+1} = f(x_n) = r x_n (1 - x_n)$$

1) $0 \leq x_n \leq 1$ and $0 \leq r \leq 4$, we have

$$0 \leq f(x_n) \leq 1$$

end pts, $x_n = 0 \Rightarrow 0$ ok

$x_n = 1 \Rightarrow 0$ ok

$$\max_{x_n} f(x_n) \Rightarrow f'(x_n) = r - 2rx_n = 0$$

$$x_{n_{\max}} = \frac{1}{2}$$

$$f(x_{\max}) = f\left(\frac{1}{2}\right) = r \frac{1}{2} \left(1 - \frac{1}{2}\right) = r \frac{1}{4}$$

$$\Rightarrow r_{\max} = 4 \Rightarrow f(x_{\max}) = 1$$

$$f''(x_n) = -2r < 0, \text{ so it is a max.}$$

3.

show 2 cycle for $s \geq 3$

$$F^2(x^*, s) = x^*$$

$$x_n(s-1-sx_n) = 0$$

$$x_{n+1} = sx_n(1-x_n) = sx_n - sx_n^2 = x_n \Rightarrow x_n = \frac{s-1}{s}$$

$$x_{n+2} = s(sx_n(1-x_n))(1-sx_n(1-x_n))$$

$$\Rightarrow x^* = x = \frac{(s^2x - s^2x^2)}{(1-sx + sx^2)}$$

$$x = \frac{s^2x - s^3x^2 + s^3x^3 - s^2x^2 + s^3x^3 - s^3x^4}{1-sx + sx^2}$$

$$0 = x(s^2-1) - x^2(s^2+s^3) + 2x^3s^3 - s^3x^4$$

we know $x=0$, and $x = \frac{s-1}{s}$ are roots.

$$0 = (s^2-1) - x(s+1)s^2 + 2x^2s^3 - s^3x^3 \div (x - \frac{s-1}{s})$$

polynomial division

$$0 = -x^2s^3 + s^2x + s^2x - s^2 - s$$

$$0 = -x^2s^2 + s^2x + sx - s - 1$$

$$0 = x^2s^2 - x(s+1)s + (s+1)$$

$$x = \frac{(r+1) \pm \sqrt{(r+1)^2 r^2 - 4r^2(r+1)}}{2r^2}$$

$$x = \frac{r+1}{2r} \pm \frac{\sqrt{r^2(r+1)(r+1-4)}}{2r^2}$$

$$x = \frac{r+1 \pm \sqrt{(r+1)(r-3)}}{2r} \quad \text{we call } x_{\pm} = q \text{ and } p$$

for real distinct roots: $0 < (r+1)(r-3)$

$$\Rightarrow 0 < r^2 - 2r - 3 \Rightarrow r = \frac{2 \pm \sqrt{4+12}}{2} \Rightarrow \cancel{5} \wedge r = 3$$

$r \text{ defined } \geq 0$

with $r = 3$ we have just one solution
for $f^{(2)}(x) = x$, other than $x = 0 \wedge x = \frac{r-1}{r}$

so $r > 3$ for $\exists$ of 2-cycle.

stability: for stability at q and p

for two cycle $x = p = f(f(x)) = f(f(p)), f(p) = q$

chain rule

$$\frac{d}{dx} [f(f(x))]_{x=p} = f'(f(p)) \cdot f'(p) = f'(q) \cdot f'(p)$$

similar for $x=q$

so $|f'(q) \cdot f'(p)|$ gives stability of 2 cycle.

$$|f'(q) \cdot f'(p)| < 1 \text{ for stable.}$$

> 1 unstable.

$$f(x) = rx(1-x) \Rightarrow f'(x) = r - 2rx$$

$$f'(q) \cdot f'(p) = (r - 2rp)(r - 2rq)$$

$$\Rightarrow |r^2 - 2r^2(q+p) + 4r^2qp| = 1$$

$$1 = |r^2(1 - 2(q+p) + 4qp)|$$

$$q+p = \frac{r(1 + \sqrt{(5+1)(r-3)})}{2r} + \frac{r(1 - \sqrt{(5+1)(r-3)})}{2r}$$

$$= \frac{\cancel{r}(5+1)}{\cancel{r}2} = \frac{5+1}{r}$$

$$qp = \left(\frac{r(1 + \sqrt{(5+1)(r-3)})}{2r} \right) \left(\frac{r(1 - \sqrt{(5+1)(r-3)})}{2r} \right)$$

$$Q P = \frac{(s+1)^2 - (s+1)(s-3)}{4s^2} = \frac{\cancel{s^2} + 2s + 1 - \cancel{s^2} + 2s + 3}{4s^2} = \frac{4(s+1)}{4s^2}$$

$$1 = \left| s^2 \left(1 - 2 \frac{s+1}{s} + 4 \left(\frac{s+1}{s^2} \right) \right) \right|$$

$$1 = | s^2 - 2(s+1)s + 4(s+1) |$$

$$1 = | -s^2 + 2s + 4 |$$

$$0 = | -s^2 + 2s + 4 | - 1$$

$$0 = -s^2 + 2s + 4 - 1 \quad \wedge \quad 0 = s^2 - 2s - 4 - 1$$

$$\left. \begin{aligned} s &= \frac{-2}{-2} \pm \sqrt{\frac{4+12}{2}} \\ s &= 1 \pm 2, \quad \Rightarrow s=3 \end{aligned} \right\} \quad \begin{aligned} s &= \frac{2 \pm \sqrt{4+20}}{2} \\ s &= 1 \pm \frac{2\sqrt{6}}{2} \quad \Rightarrow s=1+\sqrt{6} \end{aligned}$$

$$3 < s < 1 + \sqrt{6}$$

$$\Rightarrow 1 < | -s^2 + 2s + 4 | \Rightarrow \text{stable}$$

$$s > 1 + \sqrt{6} \Rightarrow \text{unstable}$$

